

Breakdown of the slowly varying envelope approximation in the self-focusing of ultrashort pulses

Jinendra K. Ranka and Alexander L. Gaeta

School of Applied and Engineering Physics, Cornell University, Ithaca, New York 14853

Received December 16, 1997

We show theoretically and experimentally that with ultrashort pulses much longer than a single optical cycle, the effects of self-steepening and space-time focusing are important for describing the nonlinear dynamics of self-focusing. Asymmetric temporal splitting of the pulse envelope is observed in which the relative magnitudes of the peaks are reversed as the input power is increased. © 1998 Optical Society of America

OCIS codes: 190.5530, 190.7110, 320.2250, 320.7121, 190.5940.

The self-focusing of ultrashort optical pulses is important for a number of processes, including Kerr-lens mode locking,¹ pulse propagation in plasmas,² and optical breakdown in bulk media.³ Recent theoretical⁴ and experimental^{5,6} results showed that the presence of even a small amount of positive group-velocity dispersion substantially alters the nonlinear dynamics of the self-focusing process. The theoretical analysis was based on the three-dimensional nonlinear Schrödinger equation (NLSE), and one of the most significant predictions of this analysis is that the pulse undergoes temporal splitting as it self-focuses in a nonlinear medium with normal group-velocity dispersion. This pulse-splitting behavior was confirmed by experiments performed in glass.^{5,6}

In deriving the NLSE, one makes the slowly varying envelope approximation (SVEA) in time, which assumes that the spectral content of the field is narrower than the carrier frequency. However, it is expected that as the pulse undergoes self-focusing, self-phase modulation can cause the initial pulse spectrum to broaden to the point that the SVEA in time is no longer well satisfied. Treatment beyond the SVEA for nonlinear pulse propagation was first considered in the plane-wave limit by inclusion of the effects of self-steepening, which in the absence of dispersion can lead to the formation of an optical shock at the trailing edge of the pulse as it propagates through the medium.^{7–10} To date, the effects of self-steepening in nonresonant media have been reported solely for one-dimensional propagation in single-mode optical fibers^{11,12} and in hollow-core waveguides filled with a noble gas.¹³ Rothenberg¹⁴ also pointed out that space-time focusing can result in the breakdown of the SVEA for pulses considerably longer than an optical cycle. In a manner similar to self-steepening, the space-time focusing contribution results in an asymmetric pulse shape with the pulse energy shifting toward the trailing edge of the pulse.

A theoretical model based on the scalar-wave equation was recently proposed¹⁵ that extends the NLSE for pulse envelopes as short as a single optical cycle. This nonlinear envelope equation (NEE) includes the effects of space-time focusing and self-steepening in a self-consistent fashion. Through numerical comparison with Maxwell equations, the NEE was shown to describe accurately the one-dimensional propagation

dynamics of a single-cycle pulse in a dispersive Kerr medium.

In this Letter we present theoretical results on self-focusing based on numerical simulations of the three-dimensional nonlinear envelope equation. Our results show for pulses that are initially much longer than a single optical cycle that the pulse-splitting dynamics can exhibit a significant deviation from the results predicted by the NLSE. These predictions are found to be in excellent agreement with experiments that we performed in bulk fused silica.

The envelope $A(\mathbf{r}, t)$ of an optical pulse centered at frequency ω propagating along the z axis in a dispersive medium with an intensity-dependent refractive index is modeled by the NEE as¹⁵

$$\frac{\partial A}{\partial z} = \frac{i}{2k} \left(1 + \frac{i}{\omega} \frac{\partial}{\partial \tau} \right)^{-1} \nabla_{\perp}^2 A - i \frac{\beta_2}{2} \frac{\partial^2 A}{\partial \tau^2} + i \frac{n_2 n_0 \omega}{2\pi} \left(1 + \frac{i}{\omega} \frac{\partial}{\partial \tau} \right) |A|^2 A, \quad (1)$$

where β_2 is the GVD, $k = n_0 \omega / c$ is the wave-vector amplitude, and $\tau = t - z/v_g$ is the retarded time for the pulse traveling at the group velocity v_g . The presence of the operator $(1 + i\partial/\omega\partial\tau)$ in the diffraction term $(\nabla_{\perp}^2 A)$ of Eq. (1) leads to space-time focusing, whereas its appearance in the nonlinear term results in self-steepening. It is apparent that for a self-consistent analysis of the self-focusing dynamics the effects of both space-time focusing and self-steepening must be included. We study Eq. (1) for the case in which the beam has radial symmetry (i.e., $\nabla_{\perp}^2 = \partial^2/\partial r^2 + \partial/r\partial r$) and assume that the input pulse is Gaussian in space and time such that $A(r, t) = A_0 \exp(-r^2/2w_0^2 - t^2/2\tau_p^2)$. The inverse of the parameter $\omega\tau_p$ is a measure of the breakdown of the SVEA for the input pulse, and the following three length scales characterize the pulse interaction: diffraction length $L_{DF} = kw_0^2/2$, dispersion length $L_{DS} = \tau_p^2/|\beta_2|$, and nonlinear length $L_{NL} = (n_0 n_2 \omega |A_0|^2 / 2\pi)^{-1}$. The ratio of L_{DF} to L_{NL} can be related to the ratio of the input power P to the critical power $P_{cr} = \tau(0.61\lambda)^2/8n_0 n_2$ through the relation $L_{DF}/L_{NL} = 0.9425P/P_{cr}$.

We numerically integrate Eq. (1) for conditions similar to those of our experiment (i.e., $L_{DF}/L_{DS} = 0.15$, $\omega\tau_p = 110$, and the length L of the medium is $2.5L_{DF}$)

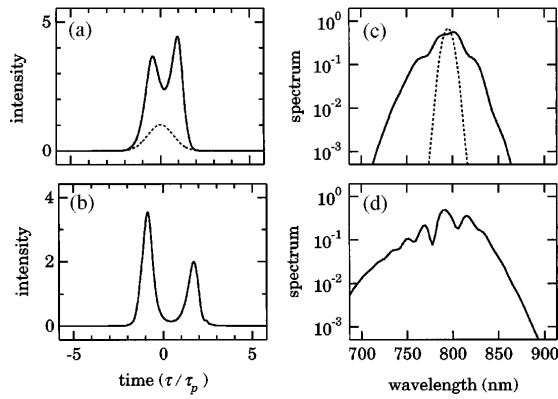


Fig. 1. Theoretical predictions for the temporal profile of (a), (b), the on-axis intensity $n_2 n_0 \omega |A|^2 L / 2\pi$ and (c), (d) the power spectra of the transmitted beam as the input power is increased. The dashed curves in (a) and (c) are the temporal profile and the spectrum of the input pulse, respectively. The peak power of the incident pulse in (a) and (c) is $1.75P_{cr}$, and that in (b) and (d) is $1.9P_{cr}$.

and examine the on-axis temporal profile, the frequency chirp, and the spectrum of the transmitted pulse (Fig. 1) as the input power is increased above the threshold for pulse splitting. Our technique for integration is based on a split-step Fourier-transform Crank–Nicholson technique in which the diffraction and the dispersion parts are calculated in frequency space and the nonlinear term is calculated in the time domain. The temporal profile and the spectrum of the input pulse are shown by the dashed curves in Figs. 1(a) and 1(c), respectively. As the input power is increased above the threshold for pulse splitting, the transmitted pulse exhibits asymmetric splitting in which the rear pulse is higher in amplitude than the front pulse [Fig. 1(a)]. This asymmetry is consistent with simulations that allowed for space–time focusing¹⁴ and with a recent theoretical analysis¹⁶ based on a set of reduced equations that also includes the effects of nonparaxiality. The corresponding pulse spectrum [Fig. 1(c)] broadens with the appearance of a blue tail, which is characteristic of the preferred generation of the blue frequency components as a result of self-steepening and space–time focusing. At higher powers ($P/P_{cr} = 1.9$), the relative amplitudes of the pulses are reversed, with the front pulse becoming more prominent [Fig. 1(b)], and the spectrum [Fig. 1(d)] exhibits additional broadening, with the blue tail becoming more pronounced. At even higher input powers, the rear pulse is further diminished relative to the front pulse. Inspection of the temporal profile of the pulse at various points in z indicates that the shift in the relative amplitudes of the split pulses is a result of the initial splitting asymmetry at threshold, in which the rear pulse is dominant. As the power is increased, the splitting occurs at decreasing values of z . At the point of the initial pulse splitting, the more-intense rear pulse has a smaller effective spot size and a broader spectrum, and as a result the effects of diffraction and dispersion diminish the peak intensity of the rear pulse more rapidly than that of the front pulse as the pulses propagate toward the output

face at $z = L$. This behavior is confirmed by inspection of the temporal profile for $P/P_{cr} = 1.9$ at the point in z at which the splitting initially occurs and by the observation that the rear pulse is more intense than the front pulse. In all our simulations the effects of self-steepening and space–time focusing made comparable contributions to the resulting asymmetric behavior. Inclusion of higher-order dispersion comparable with that of fused silica is found to have a negligible effect in our simulations.

We verified these predictions of the NEE by characterizing the on-axis temporal and spectral profiles of an ultrashort pulse transmitted through a fused-silica glass substrate as the peak power is increased above the threshold power for pulse splitting. The 78-fs pulses were produced by a Ti:sapphire regenerative laser system (Clark–MXR) operating at 795 nm. A small fraction of the output was spatially filtered, which resulted in a spatial profile that was nearly Gaussian. The beam was focused at a $75\text{-}\mu\text{m}$ spot size at the front face of a 3-cm-long fused-silica glass sample. For fused silica, $\beta_2 = 385\text{ fs}^2/\text{cm}$, and $n_2 = 2 \times 10^{-16}\text{ cm}^2/\text{W}$, which corresponds to a critical power $P_{cr} = 3\text{ MW}$. We characterized the temporal behavior by measuring the intensity cross correlation of the transmitted pulse with the initial 78-fs input pulse, which allowed for the measurement of the relative amplitudes of the split pulses. Our scanning ($\sim 1\text{-s}$ -long scan) noncollinear cross correlator incorporated a GaAsP photodiode as the nonlinear medium¹⁷ rather than a second-harmonic-crystal–detector combination typically used in such measurements. The spectrum of the transmitted beam was taken concurrently by use of a fiber-coupled spectrometer with 0.3-nm resolution.

The cross correlation and the corresponding spectrum of the transmitted pulse on axis are shown in Fig. 2 as the incident power was increased above the threshold power for pulse splitting. The dashed curves in Figs. 2(a) and 2(c) represent the autocorrelation and spectrum of the input pulse, respectively. The initial splitting threshold occurs at $P = 4.2\text{ MW}$, where the cross correlation indicates a temporal asymmetry with the energy shifted to the trailing edge. At a power of $P = 5.1\text{ MW}$ ($P/P_{cr} = 1.7$), the splitting and the asymmetry are distinct [Fig. 2(a)], and the trailing pulse has a higher amplitude. The transmitted pulse spectrum [Fig. 2(c)] exhibits significant broadening with a long blue tail. At slightly higher input powers ($P = 5.9\text{ MW}$, $P/P_{cr} = 1.97$), the asymmetry in the splitting is reversed [Fig. 2(b)], and the leading pulse becomes more pronounced. The corresponding spectrum [Fig. 2(d)] is broadened further, and the blue tail becomes more prominent. At still higher powers, the amplitude of the rear pulse becomes progressively weaker than the front pulse.

Inspection of Figs. 1 and 2 indicates that our experimental results are well described by our theoretical simulations based on the NEE. We find the agreement in the shape of the spectral wings to be particularly notable and indicative of the role that self-steepening and space–time focusing play in this

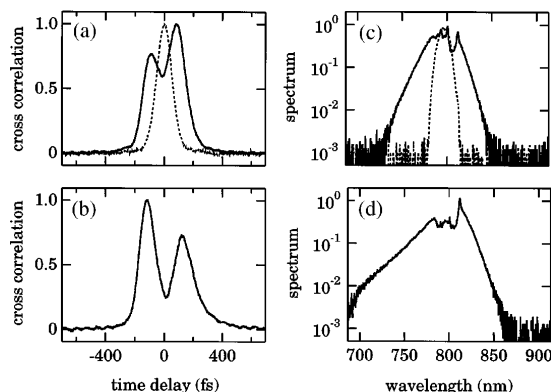


Fig. 2. Experimentally measured (a), (b) cross correlation and (c), (d) power spectra of the pulses transmitted through a 3-cm fused-silica glass sample as the input power is increased. The peak power for (a) and (c) is 5.1 MW, and that for (b) and (d) is 5.9 MW. The dashed curves in (a) and (c) are the autocorrelation and the spectrum of the input pulse, respectively.

interaction. Autocorrelation measurements of the transmitted pulse are consistent with the theoretical prediction that the split pulses are considerably narrower than the input. In our experiments we also confirmed above the threshold for pulse splitting that the rear (front) pulse is blueshifted (redshifted) with respect to the input pulse by inserting a long-wavelength-pass filter with an absorption edge at the initial wavelength of the input pulse and observing that the front (rear) pulse is nearly completely transmitted (absorbed) by the filter. We verified that the rear pulse just above the splitting threshold is always dominant by comparing the relative intensities of the split pulses for different length samples. For example, we found at input powers (e.g., $P = 6$ MW) above the threshold for pulse splitting that the front pulse was dominant in the longer sample ($L = 3$ cm), whereas the rear pulse was more pronounced in the shorter sample ($L = 1.27$ cm).

To investigate the effects of multiphoton ionization and plasma generation on the self-focusing process in our experiments, we measured the transmitted pulse energy as the peak power was increased above the critical power for splitting. In all cases shown here, we find that greater than 95% of the pulse energy is transmitted, which is consistent with the finding that the initial intensities are approximately 2 orders of magnitude smaller than the surface breakdown threshold.^{18,19} We further examined the increase in the pulse-splitting threshold as the polarization of the input pulse changed from linear to circular. Since the cross section for multiphoton absorption is much less for circular polarization than that for linear polarization,¹⁹ a modification in the pulse-splitting dynamics might be expected if nonlinear absorption played a vital role. Nevertheless, we found that the threshold increases by precisely a factor of 1.5, as expected from the decrease in n_2 by 2/3 for circularly polarized light. Theoretical simulations by us and others²⁰ that have included the effects of multiphoton ionization and plasma

formation indicate that these processes would result in an initial asymmetric splitting that is opposite the predictions of the NEE and our experimental results.

In conclusion, our theoretical and experimental results demonstrate that the nonlinear envelope equation, which includes the effects of self-steepening and space-time focusing, gives an accurate description of the pulse-splitting dynamics in the self-focusing of femtosecond laser pulses.

This research was supported through National Science Foundation grants PHY-9722603 and ECS-9612255.

References

1. I. P. Christov, V. D. Stoev, M. M. Murnane, and H. C. Kapteyn, *Opt. Lett.* **20**, 2111 (1995).
2. P. Sprangle, E. Esarey, and B. Hafizi, *Phys. Rev. Lett.* **79**, 1046 (1997).
3. D. von der Linde and H. Schüller, *J. Opt. Soc. Am. B* **13**, 216 (1996).
4. N. Zharova, A. Litvak, T. Petrova, A. Sergeev, and A. Yunakovsky, *Pis'ma Zh. Eksp. Teor. Fiz.* **44**, 12 (1986) [*JETP Lett.* **44**, 13 (1986)]; J. E. Rothenberg, *Opt. Lett.* **17**, 583 (1992); P. Chernev and V. Petrov, *Opt. Lett.* **17**, 172 (1992); G. G. Luther, A. C. Newell, J. V. Moloney, and E. M. Wright, *Opt. Lett.* **19**, 789 (1994); G. Fibich, V. M. Malkin, and G. C. Papanicolaou, *Phys. Rev. A* **52**, 4218 (1995); L. Bergé, J. J. Rasmussen, E. A. Kuznetsov, E. Shapiro, and S. K. Turitsyn, *J. Opt. Soc. Am. B* **20**, 1879 (1996).
5. J. K. Ranka, R. W. Schirmer, and A. L. Gaeta, *Phys. Rev. Lett.* **77**, 3783 (1996).
6. S. A. Diddams, H. K. Eaton, A. A. Zozulya, and T. S. Clement, *Opt. Lett.* **23**, 379 (1998).
7. F. DeMartini, C. H. Townes, T. K. Gustafson, and P. L. Kelly, *Phys. Rev.* **164**, 312 (1967).
8. G. Yang and Y. R. Shen, *Opt. Lett.* **9**, 510 (1984).
9. G. P. Agarwal, *Nonlinear Fiber Optics*, 2nd ed. (Academic, New York, 1994).
10. N. Tozar and M. Jain, *Phys. Rev. A* **23**, 1266 (1981).
11. H. Nakatsuka, D. Grischkowsky, and A. C. Balant, *Phys. Rev. Lett.* **47**, 910 (1981).
12. J. E. Rothenberg and D. Grischkowsky, *Phys. Rev. Lett.* **62**, 531 (1989).
13. M. Nisoli, S. Stagira, S. De Silvestri, O. Svelto, S. Sartania, Z. Cheng, M. Lenzer, Ch. Spielmann, and F. Krausz, *Appl. Phys. B* **65**, 189 (1997).
14. J. E. Rothenberg, *Opt. Lett.* **17**, 1340 (1992).
15. T. Brabec and F. Krausz, *Phys. Rev. Lett.* **78**, 3283 (1997).
16. G. Fibich and G. C. Papanicolaou, *Opt. Lett.* **22**, 1379 (1997).
17. J. K. Ranka, A. L. Gaeta, A. Baltuska, M. Pshenichnikov, and D. Wiersma, *Opt. Lett.* **22**, 1344 (1997).
18. A. Braun, G. Korn, X. Liu, D. Du, J. Squier, and G. Mourou, *Appl. Phys. Lett.* **20**, 73 (1995); B. C. Stuart, M. D. Feit, S. Herman, A. M. Rubenchik, B. W. Shore, and M. D. Perry, *J. Opt. Soc. Am. B* **13**, 459 (1996).
19. W. Kautek, J. Krüger, M. Lenzer, S. Sartania, C. Spielmann, and F. Krausz, *Appl. Phys. Lett.* **69**, 3146 (1996).
20. Q. Feng, J. V. Moloney, A. C. Newell, E. M. Wright, K. Cook, P. K. Kennedy, D. X. Hammer, B. A. Rockwell, and C. R. Thompson, *IEEE J. Quantum Electron.* **33**, 127 (1997).